

Linear Models for Classification Tasks

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Shortcomings of Linear Regression

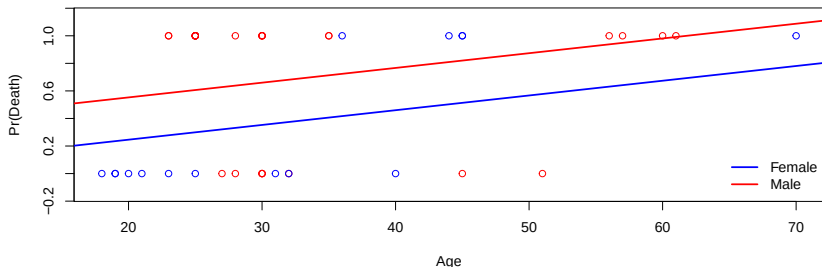
We previously introduced linear regression as a way to model the relationship between a set of predictors:

$$Y = w_0 + w_1X_1 + w_2X_2 + \dots + w_pX_p + \epsilon$$

- ▶ When Y is a binary variable, this model is problematic because its predicted values can fall outside of $[0, 1]$

Example (Donner Party Survival)

This model contains two predictors, “Age”, and “Sex” (expressed using binary one-hot encoding since this is data from 1846):



The model predicts males aged 60+ have more than a 100% probability of death, for males aged 70+ its 120%

Generalized Linear Models

- ▶ **Generalized Linear Models** provide the theoretical framework for adapting the basic structure of linear regression to classification tasks
 - ▶ To begin, linear regression can be viewed as the model:

$$Y \sim N(z, \sigma), \text{ where: } z = w_0 + w_1X_1 + w_2X_2 + \dots$$

- ▶ In this model, two components are clearly displayed:
 - ▶ A *linear predictor*, z (called a prediction score by data scientists)
 - ▶ A probability model that explains some of the variability in the outcome

Logistic Regression

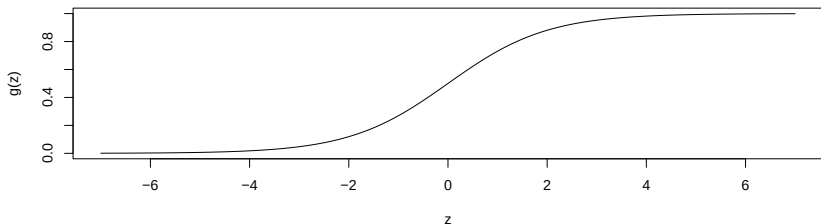
- ▶ The Normal distribution isn't suitable for a binary outcome, but the *Bernoulli distribution* is:

$$Y_i \sim \text{Ber}(g(z_i))$$

- ▶ The mean parameter of a Bernoulli distribution is $Pr(Y = 1)$
 - ▶ So, we must transform our linear predictors using a function, $g()$, such that z_i , which could be any real number is mapped to values between 0 and 1

Logistic Regression

Logistic regression is a generalized linear model that uses the *Bernoulli distribution* and the **sigmoid function**: $g(z_i) = \frac{1}{1+e^{-z_i}}$



The observed outcome (ie: $y_i = 0$ or $y_i = 1$) is a realized value from a Bernoulli distribution with a mean of $g(z_i)$

Parameters and Cost Function

The cost function used to estimate the weights of a logistic regression model is **cross-entropy loss**:

$$\text{Cost} = -\frac{1}{n} \sum_{i=1}^n (y_i \log(g(z_i)) + (1 - y_i) \log(1 - g(z_i)))$$

Or, more intuitively:

$$\text{Cost}_i = -\frac{1}{n} \log(g(z_i)) \quad \text{if } y_i = 1$$

$$\text{Cost}_i = -\frac{1}{n} \log(1 - g(z_i)) \quad \text{if } y_i = 0$$

- ▶ For observations in the positive class, higher z values correspond with lower costs
 - ▶ As z_i increases, $g(z_i) \rightarrow 1$ and $\log(1) = 0$

Optimization

The cross-entropy cost function doesn't have a closed form solution. For simplicity, we'll consider *only one stochastic gradient descent*:

$$\frac{\partial \text{Cost}}{\partial \mathbf{w}} = -y_i \mathbf{x}_i \left(\frac{g(\mathbf{x}_i)(1-g(\mathbf{x}_i))}{g(\mathbf{x}_i)} \right) + (1 - y_i) \mathbf{x}_i \left(\frac{g(\mathbf{x}_i)(1-g(\mathbf{x}_i))}{1-g(\mathbf{x}_i)} \right)$$

Note that by chain rule: $\nabla \log(g(z_i)) = \frac{1}{g(z_i)} * \frac{\partial g}{\partial z_i} * \frac{\partial z_i}{\partial \mathbf{w}}$

- ▶ $\frac{1}{g(z_i)}$ is the derivative of $\log(g(z_i))$ with respect to $g(z_i)$
- ▶ $\frac{\partial g}{\partial z_i} = g(\mathbf{x}_i)(1 - g(\mathbf{x}_i))$
- ▶ $\frac{\partial z_i}{\partial \mathbf{w}} = \mathbf{x}_i$

These components of the second term are found similarly.

Stochastic Gradient Descent Algorithm

Skipping a fair bit of algebra, the gradient function for only one stochastic gradient descent simplifies to:

$$(g(z_i) - y)\mathbf{x}_i$$

Leading to the following update scheme:

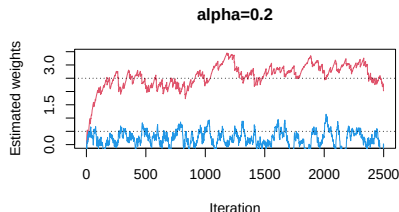
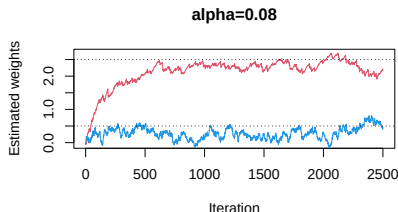
$$\mathbf{w}^{(j)} = \mathbf{w}^{(j-1)} - \alpha * (g(z_i^{(j-1)}) - y)\mathbf{x}_i$$

Noting that the prediction score, $z_i^{(j-1)}$, involves the weights from the previous iteration, $\mathbf{w}^{(j-1)}$

Optimization

The examples below demonstrate only one stochastic gradient descent on 100 data-points generated such that:

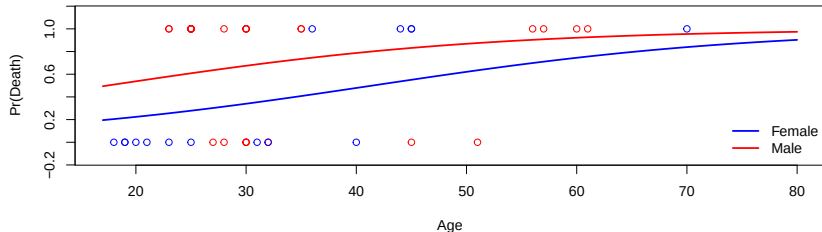
$$Y \sim \text{Ber}\left(\frac{1}{1+\exp(-(0.5+2.5x_1))}\right)$$



Visualizing Logistic Regression

Below is what the fitted logistic regression model looks like for the Donner Party example,

$$\log\left(\frac{p}{1-p}\right) = -2.40022 + 0.05788 \cdot \text{age} + 1.39310 \cdot \mathbb{1}(\text{male})$$



Visualizing Logistic Regression

The most important take-away is that our model follows a defined parametric structure, and always yields predicted probabilities between 0 and 1.

Also, when does a woman have 50/50 chance of survival/death?

- ▶ $p = 0.5$ when

$$\log\left(\frac{p}{1-p}\right) = 0$$

- ▶ for women the logistic model is

$$\log\left(\frac{p}{1-p}\right) = -2.40022 + 0.05788 \cdot \text{age}$$

- ▶ We solve for age when the logistic model is 0:

$$\text{age} = 41.5$$

Softmax Regression

- ▶ Logistic regression is designed for binary outcomes; however, the method can be generalized to multi-label classification
 - ▶ **Softmax regression**, also known as multinomial logistic regression, models the probability of class membership for each class via:

$$Pr(y_i = K) = \frac{\exp(\mathbf{w}_K^T \mathbf{x}_i)}{\sum_{l=1}^{N_k} \exp(\mathbf{w}_l^T \mathbf{x}_i)}$$

- ▶ The cost function for softmax regression is:

$$\text{Cost} = - \sum_{i=1}^n \sum_{l=1}^k \mathbb{1}(y_i = l) * \log \left(\frac{\exp(\mathbf{w}_l^T \mathbf{x}_i)}{\sum_{l=1}^k \exp(\mathbf{w}_l^T \mathbf{x}_i)} \right)$$

- ▶ For $k = 2$, this simplifies to the cross-entropy cost function of logistic regression

SKIPPED: Softmax Regression

- ▶ Softmax regression is unusual in the sense that uses a redundant set of parameters
 - ▶ That is, there are a set of weights for each of the k classes, but the same predictions could be obtained with weights for $k - 1$ classes
 - ▶ This is evident when comparing the method with logistic regression, where $k = 2$ but only 1 set of weights is estimated

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 - ▶ This is evident when comparing the method with logistic regression, where $k = 2$ but only 1 set of weights is estimated
- ▶ For this reason, there's little value in studying the weights of a softmax regression model, as there are multiple sets of weights that also will optimize the cost function
 - ▶ This is in contrast to logistic regression, where the exponentiation of a weight reflects the multiplicative impact of a 1-unit change in that variable on the odds of outcome belonging to the positive class

For more on Softmax Regression: <http://deeplearning.stanford.edu/tutorial/supervised/SoftmaxRegression/>